AquaTrace

Team Task Distribution & Development Responsibilities

Project: Community-Verified Water Pipe Repair Tracker
Team size: 8 members
Technology stack: Flutter (Dart), Firebase, Cloud Firestore, Firebase Authentication, Firebase Storage, Firebase Cloud Messaging and OpenStreetMap.

1. Team Responsibilities

Member 1 – Project Foundation & Authentication

Main responsibility: Establish the project structure and initial development environment.

- Create the GitHub repository.
- Set up the Flutter project.
- Establish the project folder and file structure.
- Configure Firebase and connect the project.
- Configure Firebase Authentication.
- Implement login and registration.
- Implement role-based authentication (Citizen, Technician and Admin).
- Create shared navigation, routing and basic reusable components.
- Prepare the README and GitHub contribution guidelines.

Deliverable: A working Flutter project foundation with Firebase integration and authentication.

Member 2 – User Profile & Settings

Main responsibility: Develop account management and application settings.

- Develop the user profile interface and profile editing.
- Implement profile picture upload.
- Implement dark/light mode.
- Implement notification preferences and location preferences.
- Implement privacy settings.
- Develop Help & Support and About AquaTrace screens.

Deliverable: Complete profile and settings functionality.

Member 3 – Fault Reporting System

Main responsibility: Allow citizens to report water pipe faults.

- Develop the fault reporting interface and fault type selection.
- Implement camera and photo capture.
- Upload images to Firebase Storage.
- Retrieve GPS coordinates.
- Implement description and severity selection.
- Save reports to Firestore and generate unique fault report IDs.

Deliverable: Complete water pipe fault reporting functionality.

Member 4 – Interactive Map & Location Services

Main responsibility: Develop the map-based interface.

- Integrate OpenStreetMap using Flutter Map.
- Display reported faults using map markers.
- Implement different marker colours for fault statuses.
- Display the current user location.
- Implement location search.
- Implement fault filtering, map zoom and navigation controls.

Deliverable: Interactive map with fault information and location services.

Member 5 – Admin Dashboard & Report Management

Main responsibility: Develop the administration dashboard and report management tools.

- Develop the admin dashboard and report statistics.
- Display pending, assigned, ongoing and completed reports.
- Implement the report review interface.
- Implement report approval and rejection.
- Implement fault severity and priority management.
- Implement technician assignment interface.
- Develop report analytics and trends.

Deliverable: Complete admin dashboard and report management functionality.

Member 6 – Technician Management System

Main responsibility: Develop the technician-side application.

- Develop the technician dashboard and assigned job queue.
- Display job details.
- Implement job acceptance and status updates.
- Implement navigation to fault locations.
- Allow technicians to upload repair photographs.
- Implement completed job history.

Deliverable: Complete technician job management functionality.

Member 7 – Community Verification & Dispute Management

Main responsibility: Implement community-based repair verification.

- Develop the repair verification interface.
- Display completed repairs awaiting verification.
- Identify nearby citizens within a defined radius.
- Implement repair confirmation.
- Implement dispute submission.
- Connect verification results to Firestore.
- Implement automatic case reopening when a repair is disputed.
- Develop dispute management functionality.

Deliverable: Complete community-based repair verification system.

Member 8 – Notifications & Report History

Main responsibility: Develop notifications and activity tracking.

- Develop the notification interface.
- Integrate Firebase Cloud Messaging.
- Notify citizens when reports are approved.
- Notify technicians about new assignments.
- Notify users about repair status changes.
- Notify admins about disputes.
- Implement notification read/unread functionality.
- Implement report activity history and status timelines.

Deliverable: Complete notification and report history system.

2. GitHub Branch Structure

Member	Suggested branch
Member 1	main / feature/project-foundation
Member 2	feature/profile-settings
Member 3	feature/fault-reporting
Member 4	feature/fault-map
Member 5	feature/admin-dashboard
Member 6	feature/technician
Member 7	feature/community-verification
Member 8	feature/notifications-history

GitHub rules

- Member 1 creates and maintains the repository.
- Each member works on their assigned feature branch.
- Commit changes regularly with clear commit messages.
- Use Pull Requests to merge features into the main branch.
- Do not directly modify another member's module without coordination.
- Test changes before merging into the main branch.

3. Shared Development Responsibilities

Activity	Responsibility
UI development	All members
Flutter development	All members
Firebase integration	Relevant module owner
Unit testing	All members
Integration testing	All members
Bug fixing	All members
Documentation	All members

Activity	Responsibility
Final presentation	All members

4. Development Rules & Integration

- **Member 1 must establish the foundation first:** create the repository, Flutter folder structure, Firebase configuration, shared navigation and initial common components.
- Before individual development begins, all members should agree on the Firestore collection structure, document fields and shared data models.
- Use common naming conventions, shared colours and application themes.
- Each member must develop and test their assigned module, including its UI and required backend integration.
- Coordinate shared dependencies and interfaces before changing common files.
- All members should participate in integration testing, bug fixing, documentation and the final demonstration.

5. Overall Module Structure

AquaTrace
■■■ Member 1: Project Foundation, Firebase Setup & Authentication
■■■ Member 2: User Profile & Settings
■■■ Member 3: Fault Reporting
■■■ Member 4: Interactive Map & Location Services
■■■ Member 5: Admin Dashboard & Report Management
■■■ Member 6: Technician Management
■■■ Member 7: Community Verification & Disputes
■■■ Member 8: Notifications & Report History

Recommended workflow: Member 1 completes and pushes the initial foundation. Once the agreed structure and shared interfaces are ready, the other seven members can develop their modules in parallel. This approach helps reduce integration conflicts.